Team 13: U.S. Customs and Border Protection (CBP) Analysis Support: Tunnel Detection System

TEAM 13 MEMBERS

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INTRODUCTION

Since 1990, more than 116 cross-border subterranean tunnels have been discovered along the continental US borders, the vast majority between US and Mexico. Tunnels present a low probability, high threat scenario to the United States and are a known means of illicit transportation of drugs, weapons, money and people across the US border. The perpetrators engaged in illicit trafficking are intelligent, tenacious, technologically innovative and they relentlessly seek to continue to expand their profitable enterprise. In today's world, confronted with the realities of terrorism and terroristic objectives, one must also acknowledge that tunnels pose a looming threat to national security. Tunnels are also a persistent military threat. A 2007 operational needs statement (ONS) from US Central Command (CENTCOM) noted that detainees were attempting to tunnel as a means to escape from the internment facilities. Another region with an emerging subterranean threat to US Forces is in Afghanistan with the Karez, or underground aquifers built to move irrigation water from mountains to villages by normal gravity-driven flow. The Karez in Afghanistan (thought to number 6,000) present the Taliban and other insurgents with a means to cache weapons and material, infiltrate and exfiltrate the battlefield and move fighters and supplies. Furthermore, in Egypt, the flow of weapons, ammunition, and other contraband under the Egyptian border has contributed significantly to the ongoing Israeli-Palestinian conflict. Open source estimates place the number of tunnels along the Israel-Gaza border between 300 and 1000. The US Army Corps of Engineers successfully answered the ONS from CENTCOM by developing and implementing the Tunnel Activity Detection System (TADS) (R2TD Implementation Directive).

The objective of the TRADOC Analysis Center - Monterey (TRAC-MTRY) study is to determine the sensor system, to include the TADS, that maximizes the probability and efficiency of detecting existing tunnels and tunnel construction activity on the US border according to geographic location, infrastructure, and historical data. We used a systems engineering and analysis approach to determine the drug traffic organization's (DTO) tunnel infiltration techniques on the Southern US border according to geographic location, infrastructure, and historical data.

This paper provides a high level overview of the systems methodology and describes the initial effort conducted during the IDFW20 to implement an agent- base approach that will enable the analysis of tunnel detection systems. The methodology was comprised of using the functional decomposition of the tunnel threat to create a set of threads and vignettes for analyzing sensor type and allocation with agent-based modeling and simulation. The tunnel threat was based on a threat enterprise model of narcotics trafficking, coupled with global information systems (GIS) data. We evaluated information instances pertinent to tunnel threat behaviors to include historical tunnel locations, urbanization of border towns and tunnel attributes to support a strategy for equipping the border with a persistent tunnel defeat capability.

APPROACH TO THE TUNNEL DETECTION AGENT-BASED MODEL.

This research coalesces exploratory data analysis, analytical models, system engineering and analysis, and modeling and simulation to gain insights into sensor allocation, configuration, placement, and prioritization of sensor field emplacement along the US southern border. We applied operational analysis research methodologies to gain insights into these issues by evaluating the data and technologies available in order to enable a well informed recommendation for the system capabilities required to detect tunnels. The goals of Team #13 were: (a) develop a preliminary design of a prototype Tunnel Detection System asset allocation/trade-off analysis vignette in an agent-based model (ABS) modeling environment, (b) develop a data farming methodology that lends itself to ease of use for analysts and (c) identify and define appropriate system measures.

DESCRIPTION OF SCENARIO

Given a generalized problem statement from the CBP, the physical domain of the problem was described and assessed. The operative effort focuses on building scenarios. Scenario building helps stakeholders make strategic decision to adapt to their several possible futures. The focus is to identify the main driving forces and areas of uncertainty.

Based on this physical domain, the threats, CBP missions, and constraints of the system were used to develop a series of scenarios (a narrative description of possible future events) each exemplified by vignettes. The vignettes (operational responses to the possible future events) were used to identify the system's actor-driven use cases. The vignettes served as the baseline for diagramming various behaviors using the systems modeling language (SySML). The use case diagrams enable initial scenario development to begin through implementation of the system's activity diagram, sequence diagrams, and functional and process decompositions, which aid in the construction of the system's concept of operations. Next, the design parameters and factor levels affecting CBP's sensor feedback and TTPs were identified. The model will be translated into an executable agent-based model, where an appropriate experimental simulation design will be applied. The results of the simulation will be captured and analyzed in order to determine an appropriate sensor outlay and which, if any, specific tunnel interdiction TTPs should be implemented by CBP.

The initial, overarching detection system use case diagram is depicted in Figure 1. The overview CBP detection system of systems includes various human elements, networked sensor systems, and key data links between the internal system and external data sources.

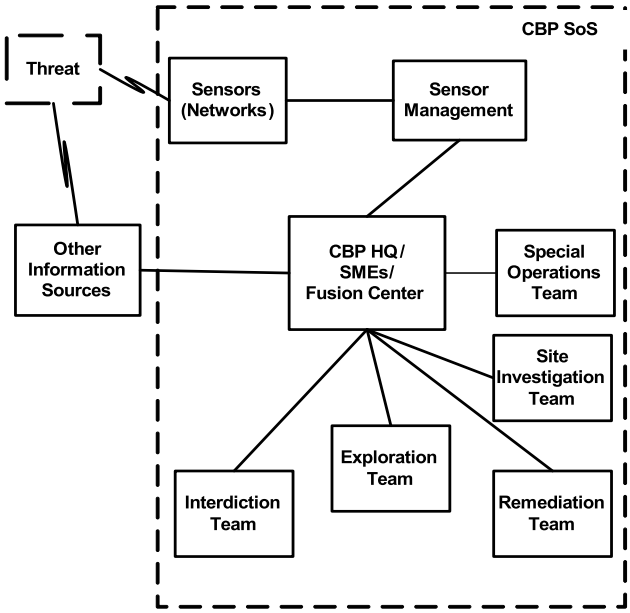

Figure 1. CBP Baseline Use Case.

After development of the baseline use case, the team applied a threat vignette (in this case, prosecution of a detected tunnel) to develop a sequence diagram depicting the process activities and functional actions of each of the relevant elements. The sequence allowed for the development of an initial system concept of operations. This CONOPS is summarized in the subsequent paragraph below.

A specific signature threshold is picked up by an array of underground sensors. The observable measurements are transmitted to the sensor management station as well as the Subject Matter Expert (SME) / Headquarters (HQ). The SME may request more information from the sensors and the management station. Concurrently, the SME may also receive external information from other sources. The sensors will respond to these requests and transmit additional parameters and observations to the SME. This data will include threat signature threshold, range, and position.

These observations allow the SME to further integrate and analyze the data and information. Once the SME has evaluated and interpreted the data to indicate that a tunnel exists, he/she will notify the Site Investigation Team to investigate the situation further. The Site Investigation Team is made up of four members. Their job will be to conduct an area site survey of the specified location. Also, they will estimate the cost of the tunnel discovery, to include determining if and where a tunnel or cavity exits. Once a tunnel is located, they will transmit their results to the SME/HQ in addition to notifying the Interdiction Team that is comprised of a Special Operations Team and Exploration Team. The Interdiction Team will investigate the specified location and drill a series of holes using a 1" – 4" auger drill bit to verify the suspected location of the tunnel. A larger bore will be implemented for ingress of robots or Special Operations teams for analysis and exploitation. They will determine the tunnel's characteristics and type which will subsequently be transmitted to the SME. The SME will direct to initiate tunnel closure actions by the Remediation Team. The team will fill the tunnel with an appropriate material (e.g. slurry substance or concrete) and will submit a tunnel closure report when complete.

In the case that the Site Investigative Team concludes that no tunnel is found, they will notify SME/HQ that no tunnel is found and flag the area and report a false alarm. The SME/HQ will develop a historical document trail for the area and note the region as a "yellow" or potential hot region for continued or increased monitoring activity.

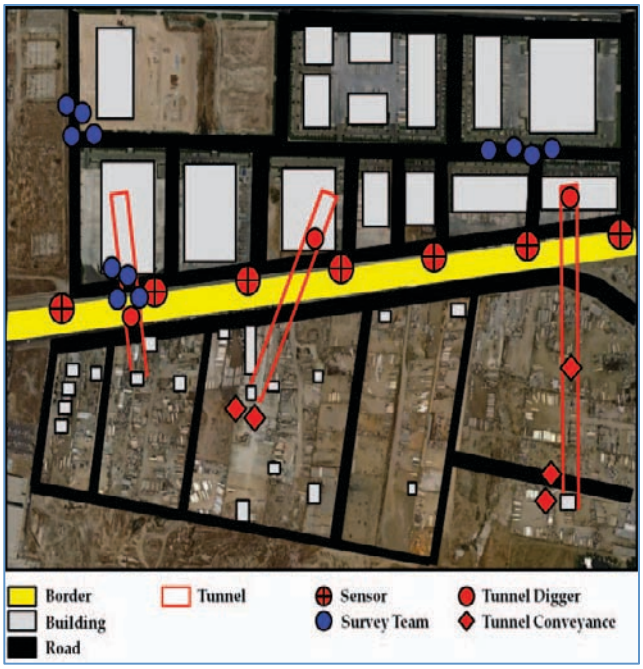

Figure 2. Agent-based Model Scenario Concept Sketch.

Based upon these CONOPs, initial efforts were made at modeling the scenario in Pythagoras (an agent-based model

that is suited for data farming, execution of large numbers of repetitions of parametric runs to identify behaviors overlaid on a dynamic landscape). A concept sketch and screenshot of the base case scenario implemented in Pythagoras are shown in Figures 2 and 3, respectively.

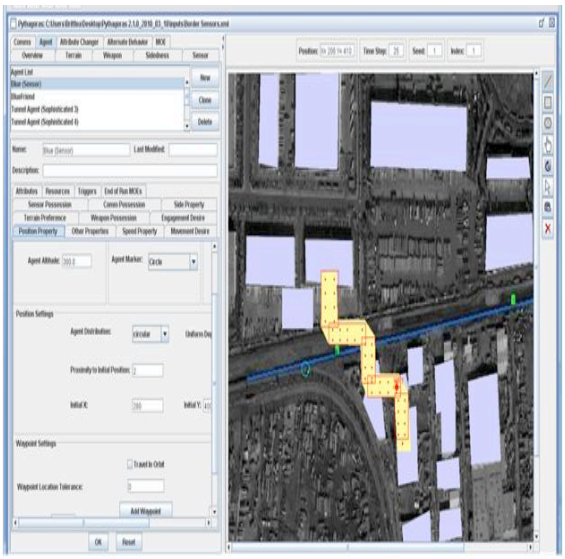

Figure 3. Agent-based Approach in Pythagoras.

Design of Experiments

The design of experiment (DOE) methodology will help us to gain insights on sensor system behavior and its interaction with key factors. Our goal is to gain such insights from factors that influence the disruption of tunnel construction and use. We will use Pythagoras, developed by Northrop Grumman, to represent tunnel detection and interdiction assets along a cross-border region of interest. During model development, we created notional factors (Table 1) that define both the characteristics and performance capabilities of the CBP sensor management system and the tunneling activities that it was designed to defeat.

Our scenario consists of a region along the United States/ Mexico border with representative road and building infrastructure as previously depicted in Figures 2 and 3. We will develop agents to represent the CBP system of systems (SoS) and threat as represented in Figure 1. Agents will include team members, vehicles, and ground sensors that encompass the SME/HQ Fusion Center, Site Investigation, Interdiction, Special Operations, Exploration, and Remediation teams, and associated sensor networks. The threat agents will consist of tunneling construction entities (digging and conveyance).

Our specific design will capture the factors relevant to the CBP SoS. Such factors include the type and number of sensors employed, sensor performance, configuration and field emplacement characteristics, the number and behavior of members in the CBP teams, as well as the characteristics of tunneling entities.

We anticipate that numerous factors consisting of multiple levels will be required to model the scenario. Employing a full factorial experiment with the 23 factors

Scenario Factors	Factor Name	Units	DOE Ranges	
			Low	High
Seismic Sensor				
Number of Sensors	NumSens	#	5	15
Probability of Detection	SensDetect	N/A	0.60	1.00
Range of Detection	SensRange	meters (m)	100	300
Target Location Error	SensTLE	N/A	0.60	1.00
Emplacement Distance	SensDistance	meters (m)	10	110
Survey Team				
Number of Survey Teams	NumSurvey	#	1	3
Movement Speed	SurveySpeed	meters/minute	60	100
Probability of Detection	SurveyDetect	N/A	0.60	1.00
Range of Detection	SurveyRange	meters	50	250
Target Location Error	SurveyTLE	N/A	0.60	1.00
Time for Survey	SurveyTime	minutes	30	90
Transition Time	SurveyTrans	minutes	30	90
Tunnel Digger				
Number of Tunnels	NumTun	#	2	6
Number of Diggers (per tunnel)	NumDig	#	1	3
Dig Rate	DigRate	meters/hour	1	5
Probability of Counter-Detection	DigDetect	N/A	0.60	1.00
Dig Delay Time	DigDelay	minutes	30	90
Tunnel Transport				
Number of Transports (per tunnel)	NumTrans	#	1	3
Transport Speed	TransSpeed	meters/minute	20	60
Probability of Counter-Detection	TransDetect	N/A	0.60	1.00
Headquarters				
Number of Headquarters	NumHQ	#	1	3
Communication Rate	CommRate	message/hour	4	60
Communication Error	CommError	N/A	0.60	1.00

Table 1. Notional Factors Representing the CBP Sensor Management System and Tunneling Activities.

presented in Table 1, each set at two levels (low & high), would result in 323 design points. A pairwise projection for the first four factors of such a design is displayed in Figure 4. Fortunately, there are more efficient alternatives to using a full factorial design.

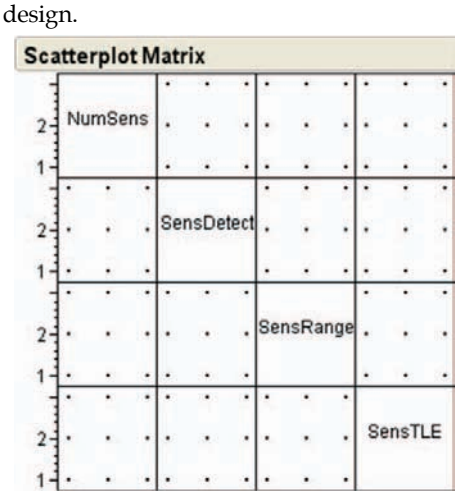

Figure 4. Scatterplot Matrix of a Factorial Design for Four Factors.

The space-filling feature of Nearly Orthogonal Latin Hypercube (NOLH) enables efficient exploration of the solution space represented by measures of effectiveness (MOEs). An NOLH experimental design will facilitate a comprehensive analysis of potential explanatory variables within our model. Rather than being restricted to two or three levels, the analyst can create a design that uses multiple levels or even a continuous range of values for each factor.

In addition to the space-filling property, orthogonal designs have no linear relationship between the regressors. The NOLH technique minimizes the correlation between factor columns, creating a nearly orthogonal design matrix. We can examine the off-diagonal elements within the correlation matrix of the design in order to measure the level of orthogonality.

Using the NOLH designs spreadsheet tool developed by Professor Susan M. Sanchez at the Naval Postgraduate School, we generated 257 design points consisting of 23 factors. Our preliminary design consists of a 257 x 23 matrix with the largest correlation of 0.089.

This design is displayed in Figure 5 which represents the space-filling, two-dimensional projections for the first four notional factors.

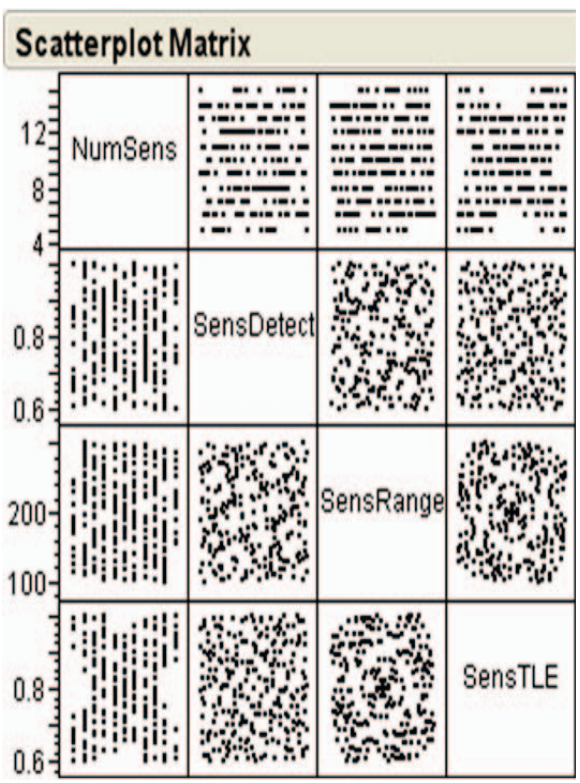

Figure 5. Scatterplot Matrix of NOLH Design for Four Factors.

SUMMARY

Analysis of Base Case

The outputs of interest consist of the possible sensor system allocations for a given CBP concept of operation. The outputs are highly multivariate, consisting of all sensor systems positions, dispositions, and strengths as well as the perception of tunnel location. That is, the output is a Common Operational Picture (COP) comprised of the adversary and/or CBP tunnel interdiction assets. The MOEs that may be derived from ABM include; time, assets and money allocation. In this manner we determine the constraints that will drive a future response.

Further Work

The challenges to tunnel defeat are attributed to the vast lengths of the border, varied geologies that degrade persistent monitoring and geophysical resolution of subsurface anomalies, as well as the unknown quantity of existing tunnels. The tunnel denial and defeat mission must be addressed in a comprehensive manner in order to be successfully implemented. This means embracing a heterogeneous system of systems approach that allows both passive and active sensing, command and control, and robotics for post discovery interrogation. Before committing to a particular technical solution, a cost-benefit study and an analysis of systems study that takes into account operational performance associated with GIS locations, risk assessment associated with GIS location, man power requirements, projected training, maintenance costs, and sustainability factors will provide the necessary oversight and will demonstrate due diligence in specifying a system solution.

Within the context of systems engineering and analysis, agent-based modeling for tunnel detection systems shows great promise as a method to enhance the evaluation of multiple detection systems under various conditions. In addition, further work may include developing and testing a DOE capability and designing a user interface appropriate to the skill level of the operator, the required response times, the resource allocations, and the operational environment. Improved automation of output analysis is also desirable, especially formatting the output reports to be more amenable for use by statistical packages for data analysis, evaluation, and assessment.

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